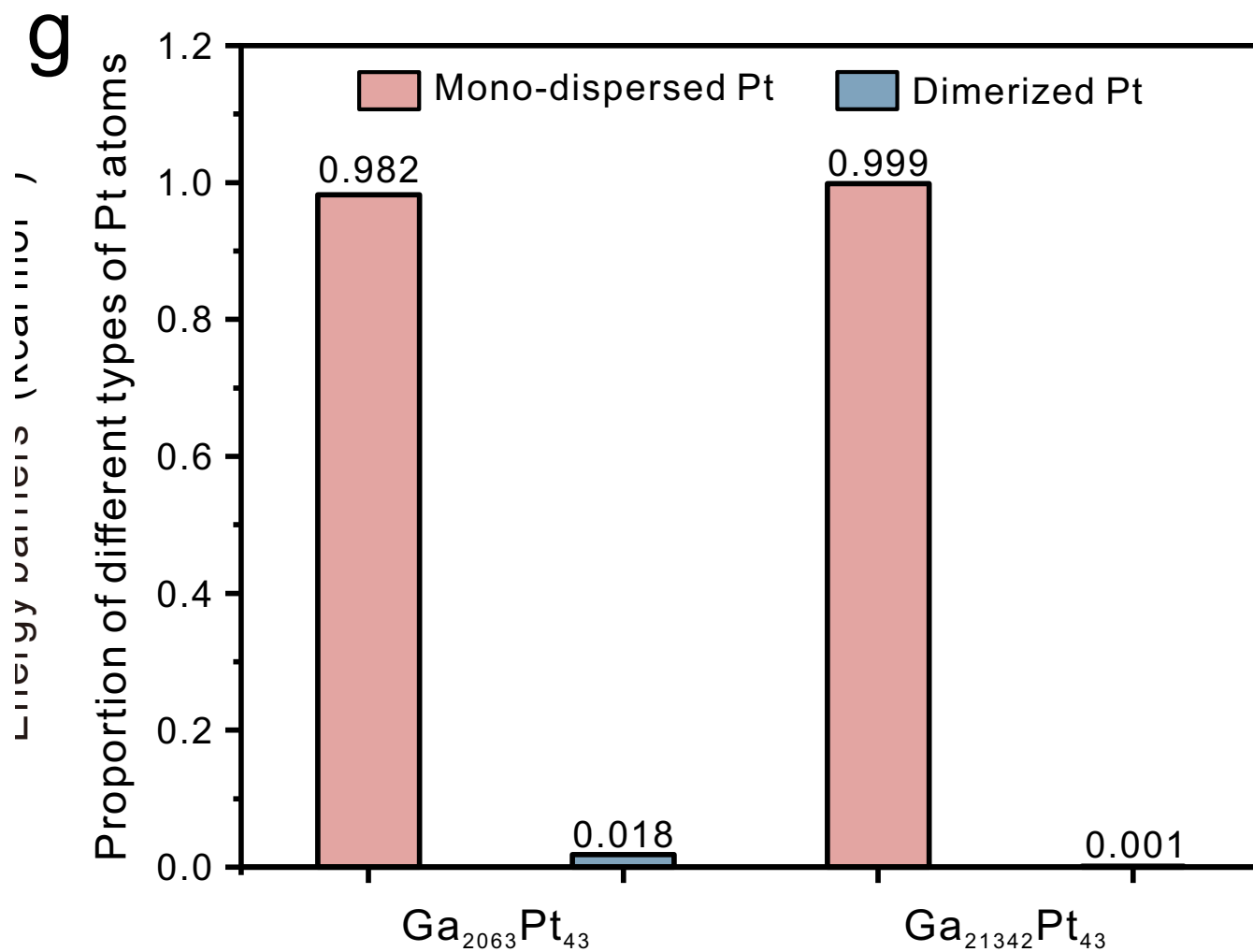


MAIN ARTICLE - REMOVED ORIGINAL CONTENT

To more intuitively demonstrate the mono-dispersion state of Pt atoms in Ga, we constructed $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures with Pt contents of 2 at% and 0.2 at%, respectively (Supplementary Fig. 14). Meanwhile, based on a training dataset containing 17347 structures and a test dataset of 1926 structures, we trained a Ga–Pt neuroevolution potential model (Supplementary Fig. 15). The consistency between the calculated and experimental solubility validates the effectiveness of the constructed machine learning force field in evaluating the dissolution of Pt in liquid Ga (Supplementary Fig. 16). Subsequently, we performed 20 ns NVT molecular dynamics (MD) simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. The simulation results clearly indicate that the Pt_{43} cluster rapidly disperses in liquid Ga (Supplementary Fig. 17). Furthermore, we analyzed the proportions of mono-dispersed and dimerized Pt states in all structures over the last 10 ns (Fig. 2g). In the $\text{Ga}_{2063}\text{Pt}_{43}$ structure, the proportion of mono-dispersed Pt is 98.2%, while in the $\text{Ga}_{21342}\text{Pt}_{43}$ structure, it reaches 99.9%. The results clearly demonstrate that in structures with Pt content of 2 at% or below, Pt atoms can be considered fully mono-dispersed at 923 K. Moreover, the calculated average numbers of

surface atoms in the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures are 5.4 and 3.0, respectively (Supplementary Figs. 18 and 19), and the temporal evolution of the vertical migration of individual Pt atoms and their probability distribution demonstrate a preferential tendency to migrate toward the surface (Supplementary Fig. 20). These indicate that a certain proportion of mono-dispersed Pt atoms can continuously exist on the liquid surface and participate in relevant catalytic reactions. Overall, the stable Ga_8Pt and similar clusters that are formed profited from the favorable affinity between Ga and Pt inhibit the aggregation and sintering of Pt during thermal motion, laying the foundation for synthesizing the highly thermal-stable SACs.



g The calculated pro-

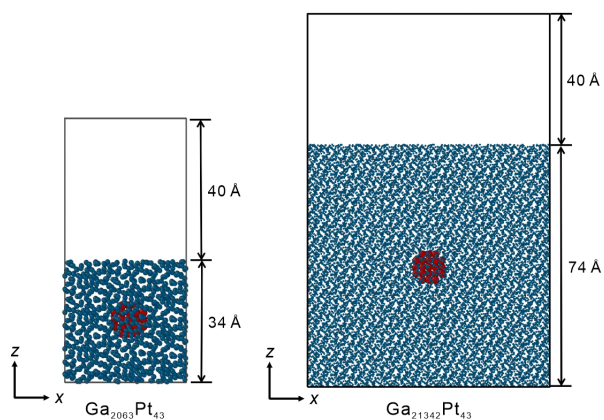
portions of mono-dispersed and dimerized Pt states in $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures (mono-dispersed Pt is defined as Pt atoms surrounded exclusively by Ga atoms within a 3.2 Å range, while dimerized Pt is defined as Pt atom that, within a 3.2 Å range, is surrounded by Ga atoms and exactly one additional Pt atom).

Ga-Pt machine learning force field training and molecular dynamics simulations. We constructed a dataset for the Ga-Pt binary system, comprising 17347 training structures and 1926 test structures. These structures cover bulk and surface structures of α -Ga, β -Ga, γ -Ga, Pt, Ga₂Pt, Ga₃Pt, Ga₃Pt₂, Ga₃Pt₅, and Ga₇Pt₃, Ga-Pt bulk structures with varying compositions (Ga₅₀Pt₁, Ga₄₉Pt₂, Ga₄₈Pt₃, Ga₄₇Pt₄, Ga₄₆Pt₅, Ga₄₅Pt₆, Ga₄₄Pt₇, Ga₄₃Pt₈, Ga₄₂Pt₉, Ga₄₁Pt₁₀, Ga₄₀Pt₁₁, Ga₃₉Pt₁₂, Ga₃₈Pt₁₃, Ga₃₇Pt₁₄, Ga₃₆Pt₁₅, Ga₃₅Pt₁₆, Ga₃₄Pt₁₇, Ga₃₃Pt₁₈, Ga₃₂Pt₁₉, and Ga₃₁Pt₂₀). Nanoparticle systems, such as Pt₆, Pt₁₃, Pt₁₉, and Pt₄₃ were also considered. The dataset was initially generated using isothermal-isobaric (*NPT*) ab initio MD simulations under ambient pressure (0.0001 GPa) and temperatures ranging from 0 to 9000 K. A machine learning potential was trained using data from AIMD simulations.

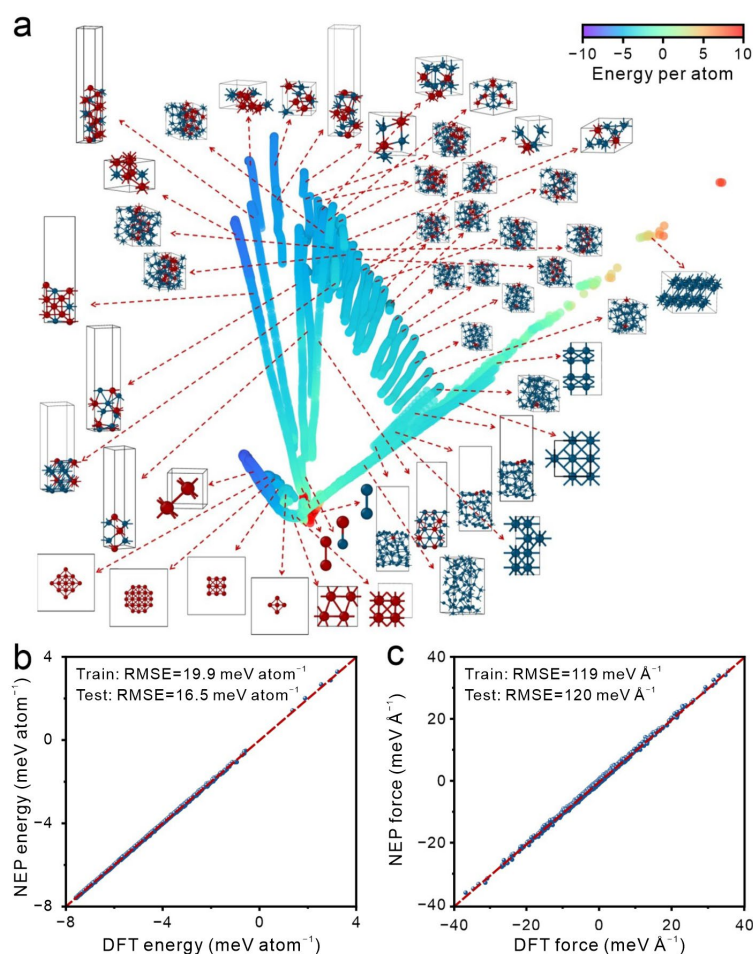
Density functional theory calculations. Density functional theory (DFT) calculations were performed using the Vienna ab initio simulation package (VASP). The calculations employed a plane-wave basis set with the projector augmented-wave method and standard pseudopotentials for Ga and Pt. The GGA-PBE_{sol} density functional was selected, along with D3 dispersion interaction corrections. All calculations were conducted in high-precision mode (PREC = accurate), with a plane-wave cutoff energy set to 400 eV and without symmetry constraints (ISYM = 0). To ensure accuracy, the electronic self-consistent loop converged to a tolerance of 10^{-6} eV (EDIFF = 1.0×10^{-6}). The Methfessel-Paxton first-order smearing method (ISMEAR = 1) was applied. The conjugate gradient algorithm was used for electronic minimization, and the maximum number of steps for electronic self-consistent (SCF) cycles was set to 999. A *k*-point mesh was automatically generated based on the reciprocal lattice vectors of the unit cell, with a *k*-point spacing of approximately $0.040 \text{ \AA}^{-1}$ in each direction. Only single-point calculations were performed, as the DFT computations were primarily used to generate labels for training data.

Machine learning force field training. The training of the force field was performed using the efficient neural evolutionary potential (NEP) method and the GPUMD 3.9.5 software. The training dataset was sourced from the previously completed PBE_{sol}-D3 calculations. The cutoff radii for the radial and angular descriptors were set to 8 and 6 Å, respectively. The weights for energy, force, and virial stress were 1.0, 1.0, and 0.1, respectively. The number of neurons in the hidden layer of the neural network (NN) was 50. The NEP potential function underwent 200,000 optimization iterations over the entire training set. The final results included the root mean square errors of energy and atomic forces.

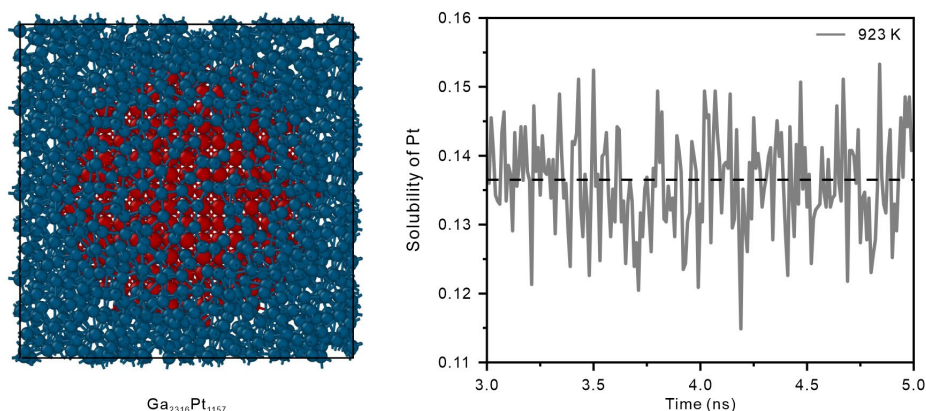
SUPPLEMENTARY INFORMATION - REMOVED ORIGINAL
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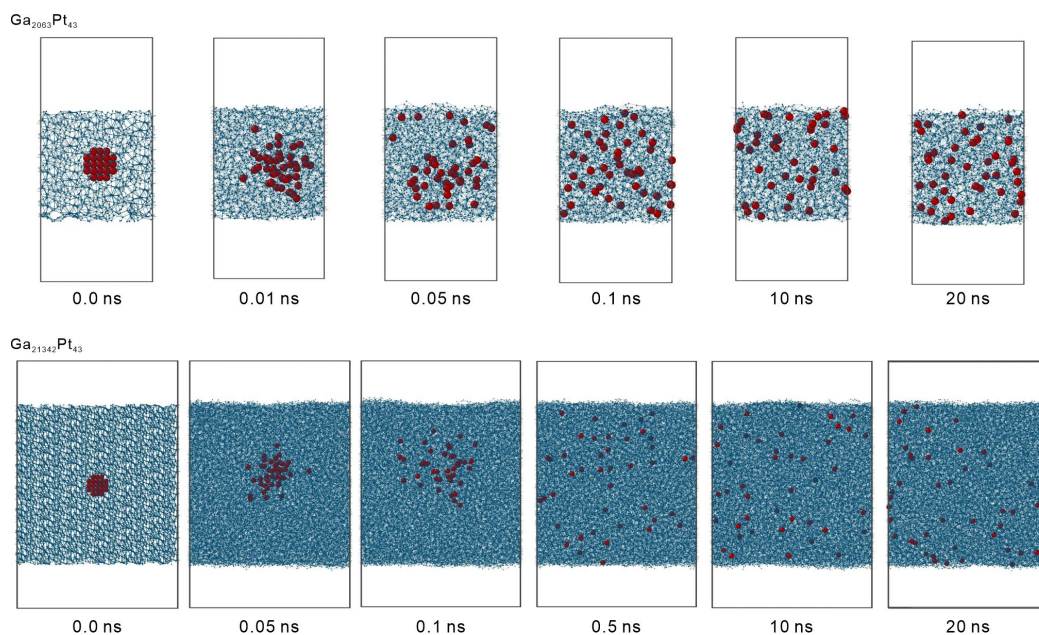
Supplementary Fig. 14 | The constructed $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures with Pt contents of 2 at% and 0.2 at%, respectively. In the subsequent catalytic process, to maximize atom economy, a liquid metal-based catalyst with 0.2 at% Pt content was selected for catalytic testing. Therefore, we also constructed a 0.2 at% Pt structure. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.



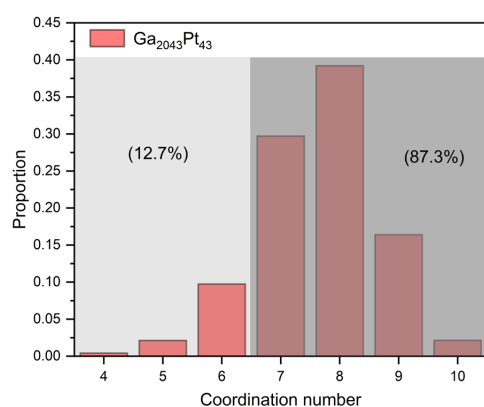
Supplementary Fig. 15 | The trained Ga–Pt neuroevolution potential model. (a) Sketch-map visualization of the Ga–Pt structural dataset. The regression plots for Ga–Pt NEP evaluated on the testing set, with an RMSE in energy (b) per atom and atomic force (c) of 16.5 meV atom⁻¹ and 120 meV Å⁻¹, respectively. A sketch-map comprises 17347 structures, with each colored point representing an individual atomic configuration (structure). The positions of the points are determined by principal component analysis of structural learning descriptors, and their colors correspond to the energy of the respective structure. The diversity of the dataset is illustrated by displaying examples of atomic configurations from different regions of the sketch map. In the crystal structures, dark blue and dark red spheres represent Ga and Pt atoms, respectively.



Supplementary Fig. 16 | The constructed Ga₂₃₁₆Pt₁₁₅₇ structure and the calculated solubility curve of Pt atoms in this structure at 923 K over time. After training, the Ga–Pt machine learning force field (MLFF) was first used to calculate the solubility of Pt in the Ga₂₃₁₆Pt₁₁₅₇ structure. We performed a 5 ns NVT ensemble simulation of the Ga₂₃₁₆Pt₁₁₅₇ structure at 923 K with a time step of 2 fs. For the solubility evaluation, the ratio of the number of Ga atoms with only Ga atoms as nearest neighbors within a 3.2 Å range to the total number of Ga atoms was statistically analyzed. The calculated average solubility of Pt in liquid Ga at 923 K was 13.6%, which is consistent with experimental results⁶. This demonstrates that the constructed machine learning force field can effectively evaluate the dissolution properties of Pt in Ga.

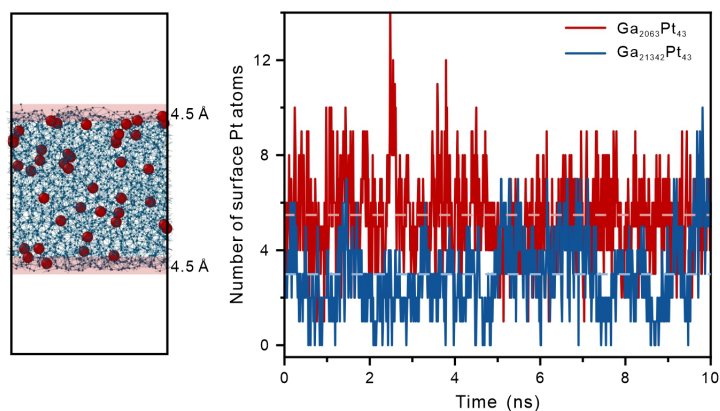


Supplementary Fig. 17 | The 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. Structural snapshots at different time points are shown with the side (xz) view. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.

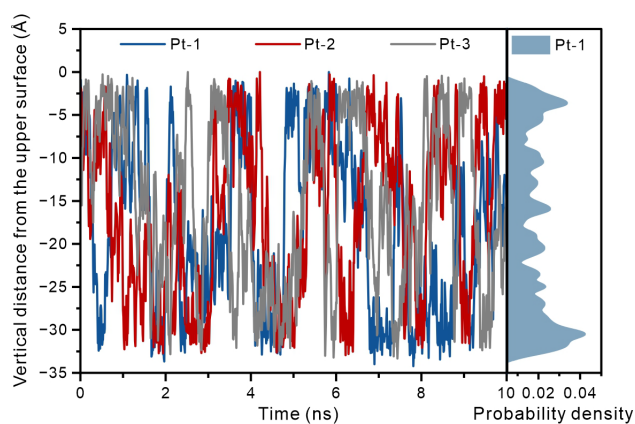


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2 **Supplementary Fig. 18 | Statistical distribution of coordination numbers for Pt atoms in**
3 **Ga₂₀₆₃Pt₄₃ from molecular dynamics simulations.** Pt atoms with coordination numbers of 6 or
4 below are considered surface Pt atoms. Calibrating the vertical height range of surface atoms based
5 on the number of surface atoms obtained from the difference in coordination numbers.



Supplementary Fig. 19 | The curves of the number of surface atoms over time in the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures, where the surface atom region is defined as ($Z_{\text{max}} - 4.5$) to Z_{max} and Z_{min} to $Z_{\text{min}} + 4.5$. The 4.5 Å threshold is determined based on the equivalence between the number of atoms counted in these regions and the number of atoms counted with a coordination number less than or equal to (the most frequent coordination number – 2) (Supplementary Fig. 18).



Supplementary Fig. 20 | The temporal evolution of the vertical migration for three randomly selected Pt atoms, and the probability density distribution of one of these Pt atoms along the vertical direction.

RESPONSE LETTER - REMOVED ORIGINAL CONTENT

Reviewer #3

(Remarks to the Author)

This work reports liquid metal dispersed single-atom catalyst by dissolving Pt into molten Ga. In fact, it is the Pt-Ga nanoalloy in previous work. I cannot make sure this kind of nanoalloy with dissolving Pt into molten Ga is the single-atom catalyst. I donot agree with the definition of molten nanoalloy as liquid metal dispersed single-atom catalyst. Thus, I donot think this work should be published in Nature Communications. Other Comments:

- 1) I suggest the authors to make sure the definition of single-atom catalyst. This kind of molten Pt-Ga nanoalloy is not new, and their catalytic mechanism is the contribution of both Pt and Ga. It is NOT the case of single-atom catalyst.
- 2) The mechanism of high temperature stability is not clear. The authors need to prepare a single Pt atom in molten Ga, and Pt almost remain the top of the molten Ga cluster and also high temperature stability.
- 3) The data analysis for liquid metal dispersed single-atom catalyst is not enough for understanding high temperature stability. How to certify that the Pt atoms donot form dimer and trimer inside the molten Ga after long-time high temperature test?

Reviewer #3

(Remarks to the Author)

I agree with Reviewer #1 that a single Pt atom can be stabilised in liquid gallium to perform high-temperature dehydrogenation catalysis is not new. The revision still lacks the required novelty for a publication in Nature Communications. I also agree with Reviewer #3 that the definition of molten nanoalloy is not liquid metal dispersed single-atom catalyst. This kind of molten Pt–Ga nanoalloy is not new, and should not be published in Nature Communications.

Firstly, the single-atom state of Pt within liquid metal was demonstrated by theoretical calculation, machine learning, and experiments.

iii) The statistical analysis dispersion state of Pt in the Pt–Ga catalyst.

Furthermore, to more intuitively demonstrate the single-atom state of Pt atoms in Ga, we trained a Ga–Pt neuroevolution potential model (which was updated in our revised manuscript as Supplementary Fig. 9). Subsequently, we performed 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. The simulation results clearly indicate that the Pt_{43} cluster rapidly disperses in liquid Ga (Fig. R3). Furthermore, we analyzed the proportions of mono-dispersed and dimerized Pt states in all structures over the last 10 ns (Fig. R4). In the $\text{Ga}_{2063}\text{Pt}_{43}$ structure, the proportion of mono-dispersed Pt was 98.2%, while in the $\text{Ga}_{21342}\text{Pt}_{43}$ structure, it reached 99.9%. The results clearly demonstrate that in structures with Pt content of 2 at% or below, Pt atoms can be considered fully mono-dispersed at 923 K. To explore the dispersion state of Pt in Ga, we expanded the calculation system and analyzed the proportion of Pt in a single-atom state and the formation of Pt clusters in a larger model. Therefore, **Pt atoms exist as single atoms and do not undergo aggregation in the Pt–Ga catalyst.**

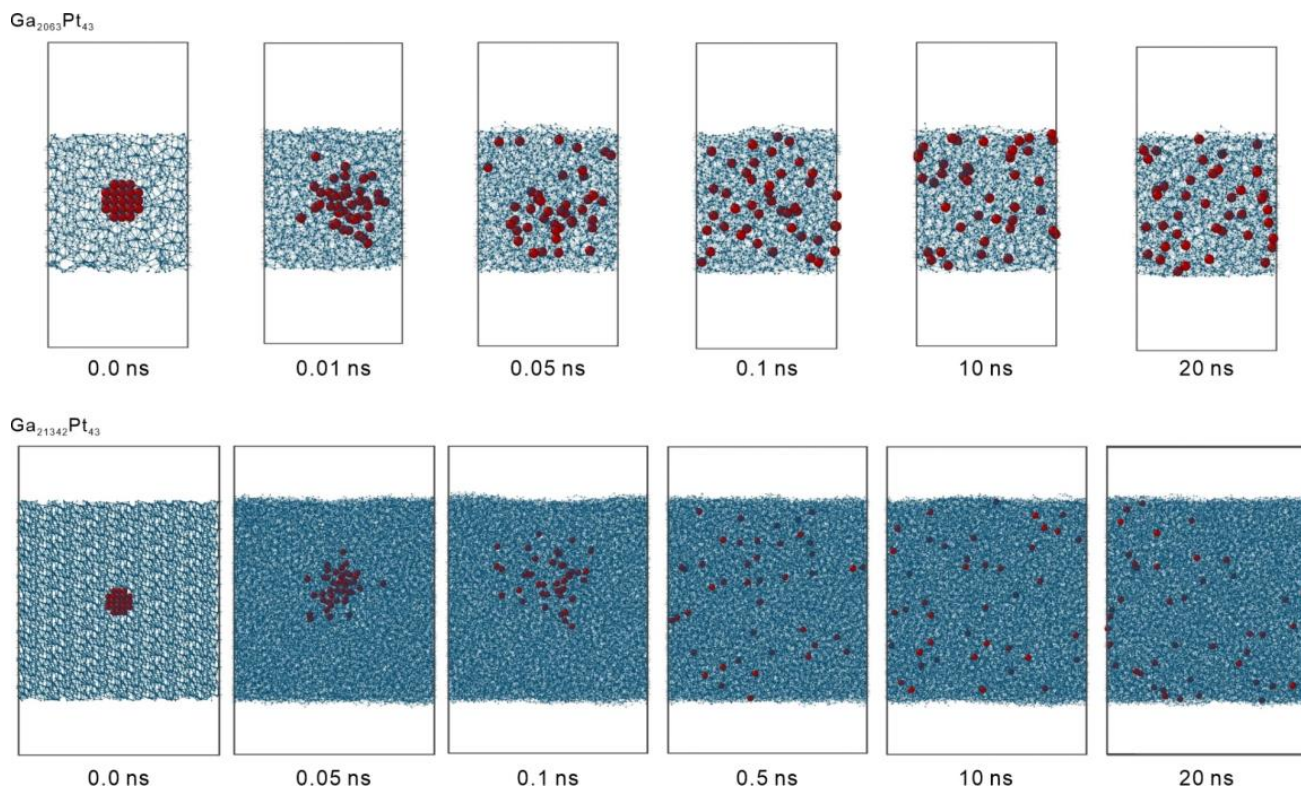


Fig. R3 | The 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. Structural snapshots at different time points are shown with the side (xz) view. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.

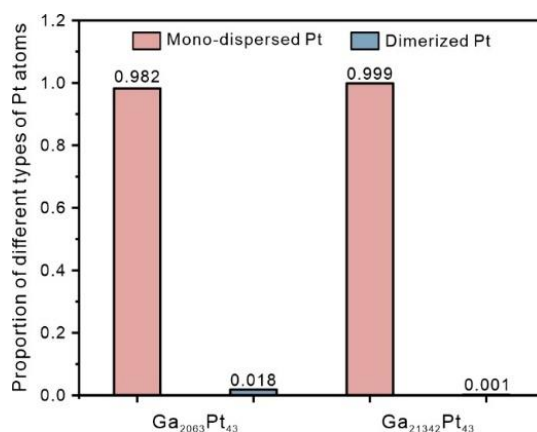


Fig. R4 | The calculated proportions of mono-dispersed and dimerized Pt states in $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures. Mono-dispersed Pt is defined as Pt atoms surrounded exclusively by Ga atoms within a 3.2 Å range, while Dimerized Pt is defined as Pt atom that, within a 3.2 Å range, is surrounded by Ga atoms and exactly one additional Pt atom.

Changes: We have added the theoretical calculations to demonstrate the single-atom state of Pt atoms within Ga more intuitively by training a Ga–Pt neuroevolution potential model in our revised manuscript. And we also verified that Pt can be stably dispersed on the Ga surface to exert catalytic activity. The changes in the revised manuscript are marked in red copy.

Page 9 in the revised manuscript:

To more intuitively demonstrate the mono-dispersion state of Pt atoms in Ga, we constructed $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures with Pt contents of 2 at% and 0.2 at%, respectively (Supplementary Fig. 8). Meanwhile, based on a training dataset containing 17347 structures and a test dataset of 1926 structures, we trained a Ga–Pt neuroevolution potential model (Supplementary Fig. 9). Subsequently, we performed 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. The simulation results clearly indicate that the Pt_{43} cluster rapidly disperse in liquid Ga (Supplementary Fig. 10). Furthermore, we analyzed the proportions of mono-dispersed and dimerized Pt states in all structures over the last 10 ns (Fig. 2g). In the $\text{Ga}_{2063}\text{Pt}_{43}$ structure, the proportion of mono-dispersed Pt was 98.2%, while in the $\text{Ga}_{21342}\text{Pt}_{43}$ structure, it reached 99.9%. The results clearly demonstrate that in structures with Pt content of 2 at% or below, Pt atoms can be considered fully mono-dispersed at 923 K. Moreover, the calculated average numbers of surface atoms in the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures are 5.4 and 3.0, respectively (Supplementary Fig. 12 and Supplementary Fig. 13), and the temporal evolution of the vertical migration of individual Pt atoms and their probability distribution demonstrate a preferential tendency to migrate toward the surface (Supplementary Fig. 14). These indicate that a certain proportion of mono-dispersed Pt atoms can continuously exist on the liquid surface and participate in relevant catalytic reactions.

Page 5–7 in the revised Supplementary Information:

Ga–Pt machine learning force field training and molecular dynamics simulations

We constructed a dataset for the Ga–Pt binary system, comprising 17347 training structures and 1926 test structures. These structures cover bulk and surface structures of

α -Ga, β -Ga, γ -Ga, Pt, Ga₂Pt, Ga₃Pt, Ga₃Pt₂, Ga₃Pt₅ and Ga₇Pt₃, Ga–Pt bulk structures with varying compositions (Ga₅₀Pt₁, Ga₄₉Pt₂, Ga₄₈Pt₃, Ga₄₇Pt₄, Ga₄₆Pt₅, Ga₄₅Pt₆, Ga₄₄Pt₇, Ga₄₃Pt₈, Ga₄₂Pt₉, Ga₄₁Pt₁₀, Ga₄₀Pt₁₁, Ga₃₉Pt₁₂, Ga₃₈Pt₁₃, Ga₃₇Pt₁₄, Ga₃₆Pt₁₅, Ga₃₅Pt₁₆, Ga₃₄Pt₁₇, Ga₃₃Pt₁₈, Ga₃₂Pt₁₉, Ga₃₁Pt₂₀). Nanoparticle systems such as Pt₆, Pt₁₃, Pt₁₉ and Pt₄₃ were also considered. The dataset was initially generated using isothermal–isobaric (NPT) Ab initio molecular dynamics (MD) simulations under ambient pressure (0.0001 GPa) and temperatures ranging from 0 to 9000 K. A machine learning potential was trained using data from ab initio molecular dynamics (AIMD) simulations.

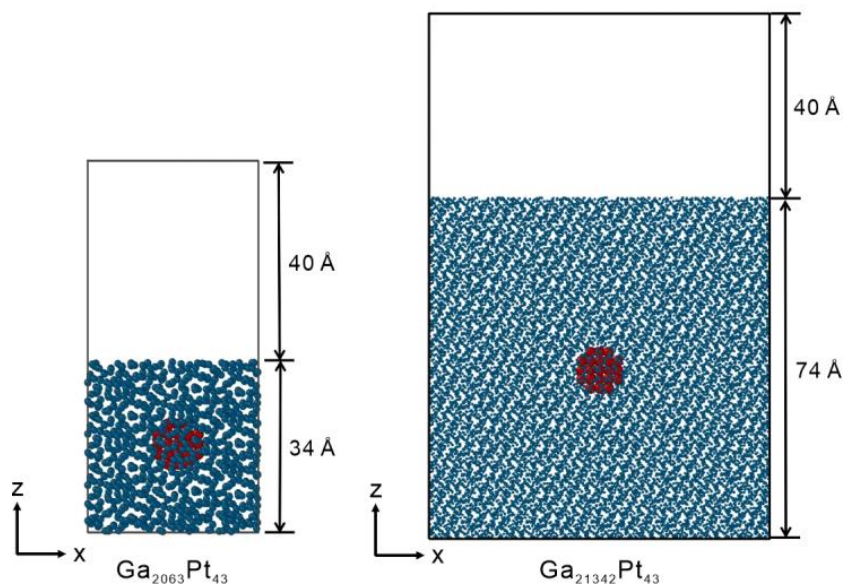
Density Functional Theory Calculations

Density functional theory (DFT) calculations were performed using the Vienna Ab initio simulation package (VASP). The calculations employed a plane-wave basis set with the projector augmented-wave (PAW) method and standard pseudo-potentials for Ga and Pt. The GGA-PBE_{sol} density functional was selected, along with D3 dispersion interaction corrections. All calculations were conducted in high-precision mode (PREC = Accurate), with a plane-wave cutoff energy set to 400 eV and without symmetry constraints (ISYM = 0). To ensure accuracy, the electronic self-consistent loop converged to a tolerance of 10^{-6} eV (EDIFF=1.0E⁻⁶). The Methfessel-Paxton first-order smearing method (ISMEAR=1) was applied. The conjugate gradient algorithm was used for electronic minimization, and the maximum number of steps for electronic self-consistent (SCF) cycles was set to 999. A *k*-point mesh was automatically generated based on the reciprocal lattice vectors of the unit cell, with a *k*-point spacing of approximately 0.040 Å⁻¹ in each direction. Only single-point calculations were performed, as the DFT computations were primarily used to generate labels for training data.

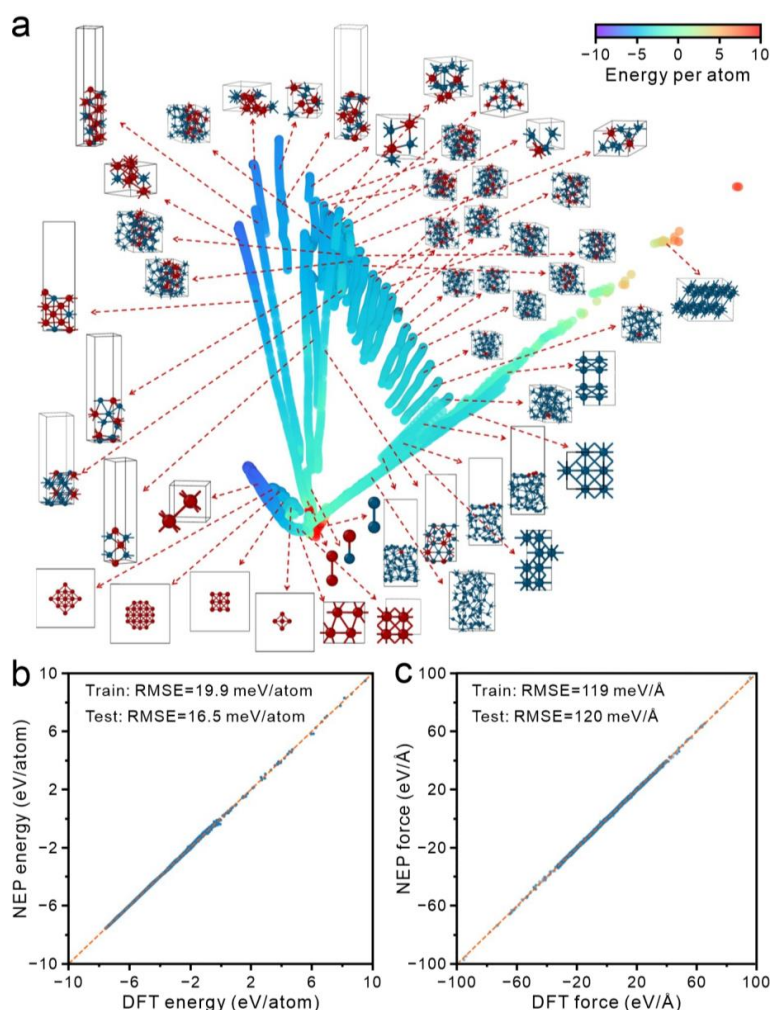
Machine Learning Force Field Training

The training of the force field is performed using the efficient Neural Evolutionary Potential (NEP) method and the GPUMD code. The training dataset is sourced from the previously completed PBE_{sol}-D3 calculations. The cutoff radii for the radial and angular descriptors are set to 8 Å and 6 Å, respectively. The weights for energy, force, and virial stress are 1.0, 1.0, and 0.1, respectively. The number of neurons in the hidden layer of the neural network (NN) is 50. The NEP potential function underwent 200000 optimization iterations over the entire training set. The final results include the root mean square errors (RMSE) of energy and atomic forces.

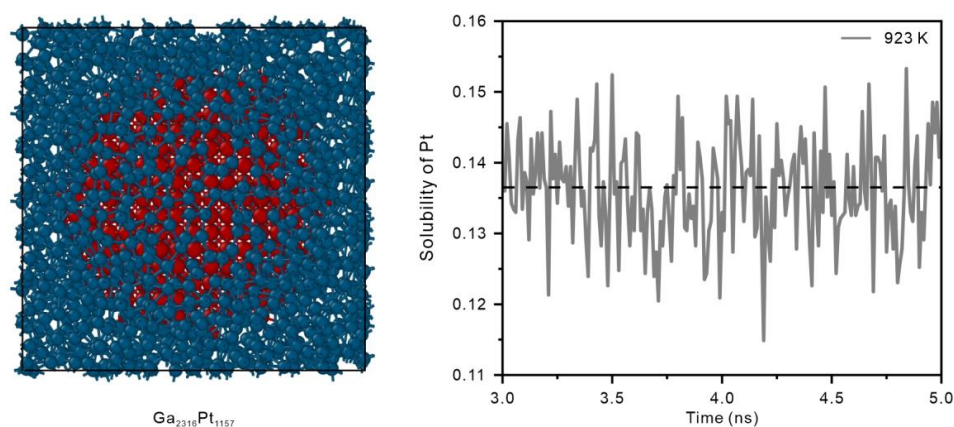
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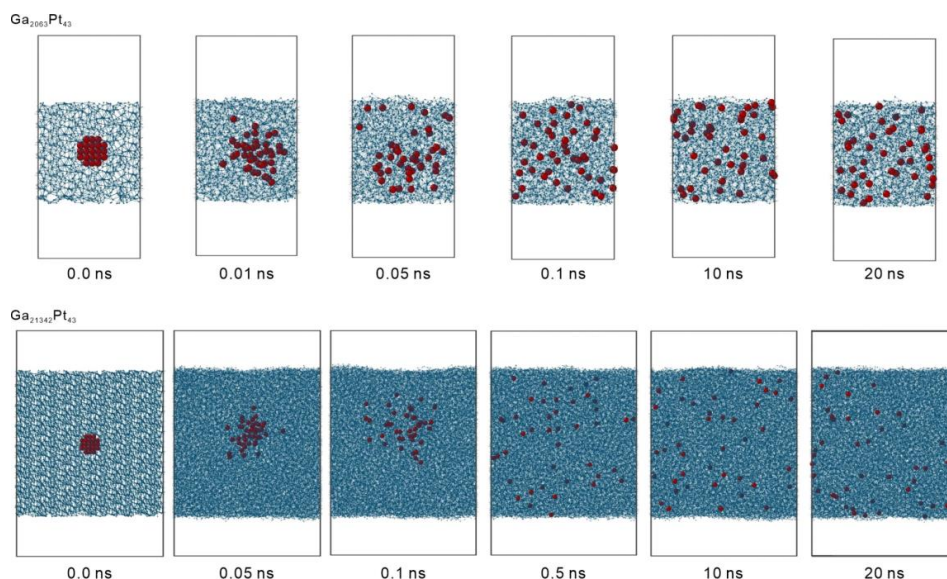
Supplementary Fig. 8 | The constructed $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures with Pt contents of 2 at% and 0.2 at%, respectively. In the subsequent catalytic process, to maximize atom economy, a liquid metal-based catalyst with 0.2 at% Pt content was selected for catalytic testing. Therefore, we also constructed a 0.2 at% Pt structure. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.



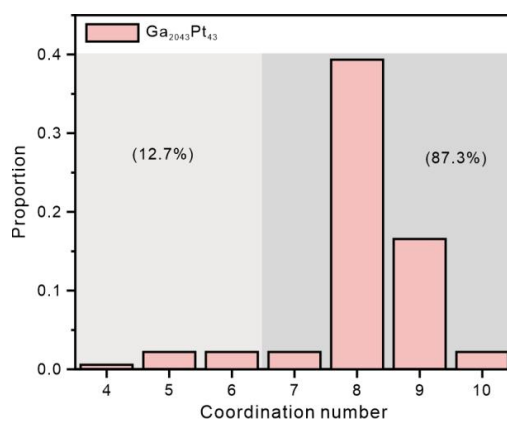
Supplementary Fig. 9 | The trained Ga-Pt neuroevolution potential model. **a** Sketch-map visualization of the Ga-Pt structural dataset. **b, c** The regression plots for Ga-Pt NEP evaluated on the training set, with an RMSE in energy (b) per atom and atomic force (c) of $19.9 \text{ meV atom}^{-1}$ and 119 meV Å^{-1} , respectively. A sketch-map comprises 17347 structures, with each colored point representing an individual atomic configuration (structure). The positions of the points are determined by principal component analysis of structural learning descriptors, and their colors correspond to the energy of the respective structure. The diversity of the dataset is illustrated by displaying examples of atomic configurations from different regions of the sketch map. In the crystal structures, dark blue and dark red spheres represent Ga and Pt atoms, respectively.



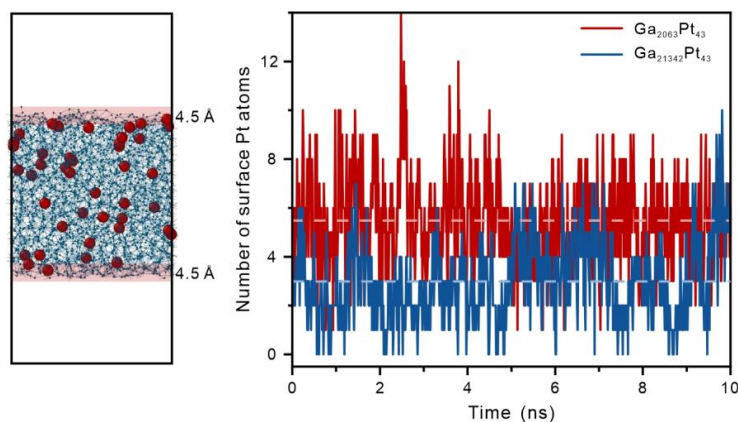
Supplementary Fig. 10 | The constructed $\text{Ga}_{2316}\text{Pt}_{1157}$ structure and the calculated solubility curve of Pt atoms in this structure at 923 K over time. After training, the Ga–Pt machine learning force field (MLFF) was first used to calculate the solubility of Pt in the $\text{Ga}_{2316}\text{Pt}_{1157}$ structure. We performed a 5 ns NVT ensemble simulation of the $\text{Ga}_{2316}\text{Pt}_{1157}$ structure at 923 K with a time step of 2 fs. For the solubility evaluation, the ratio of the number of Ga atoms with only Ga atoms as nearest neighbors within a 3.2 Å range to the total number of Ga atoms was statistically analyzed. The calculated average solubility of Pt in liquid Ga at 923 K was 13.6%, which is consistent with experimental results⁸. This demonstrates that the constructed machine learning force field can effectively evaluate the dissolution properties of Pt in Ga.



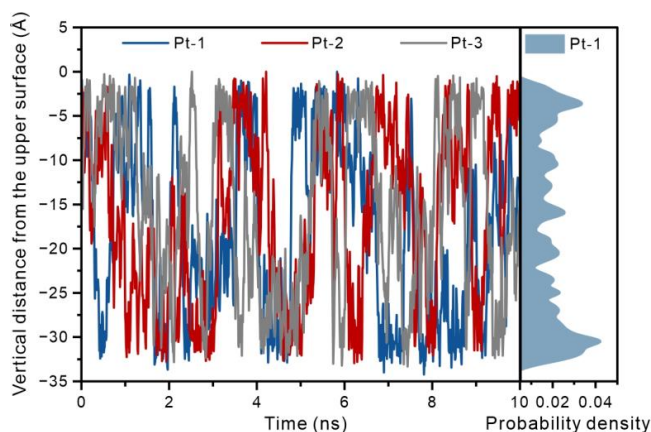
Supplementary Fig. 11 | The 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. Structural snapshots at different time points are shown with the side (xz) view. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.



Supplementary Fig. 12 | Statistical distribution of coordination numbers for Pt atoms in $\text{Ga}_{2063}\text{Pt}_{43}$ from molecular dynamics simulations. Pt atoms with coordination numbers of 6 or below are considered surface Pt atoms. Calibrating the vertical height range of surface atoms based on the number of surface atoms obtained from the difference in coordination numbers.



Supplementary Fig. 13 | The curves of the number of surface atoms over time in the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures, where the surface atom region is defined as $(Z_{\text{max}} - 4.5)$ to Z_{max} and Z_{min} to $Z_{\text{min}} + 4.5$. The 4.5 Å threshold is determined based on the equivalence between the number of atoms counted in these regions and the number of atoms counted with a coordination number less than or equal to (the most frequent coordination number – 2) (Supplementary Fig. 12).



Supplementary Fig. 14 | The temporal evolution of the vertical migration for three randomly selected Pt atoms, and the probability density distribution of one of these Pt atoms along the vertical direction.

and machine learning.

c) The statistical analysis of the dispersion state of Pt in the Pt–Ga catalyst. Furthermore, to more intuitively demonstrate the single-atom state of Pt atoms in Ga, we trained a Ga–Pt neuroevolution potential model (which was updated in our revised manuscript as Supplementary Fig. 9). Subsequently, we performed 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. The simulation results clearly indicate that the Pt_{43} cluster rapidly disperses in liquid Ga (Fig. R14). Furthermore, we analyzed the proportions of mono-dispersed and dimerized Pt states in all structures over the last 10 ns (Fig. R15). In the $\text{Ga}_{2063}\text{Pt}_{43}$ structure, the proportion of mono-dispersed Pt was 98.2%, while in the $\text{Ga}_{21342}\text{Pt}_{43}$ structure, it reached 99.9%. The results clearly demonstrate that in structures with Pt content of 2 at% or below, Pt atoms can be considered fully mono-dispersed at 923 K. To explore the dispersion state of Pt in Ga, we expanded the calculation system and analyzed the proportion of Pt in a single-atom state and the formation of Pt clusters in a larger model. Therefore, **Pt atoms exist as single atoms and do not undergo aggregation in the Pt–Ga catalyst.**

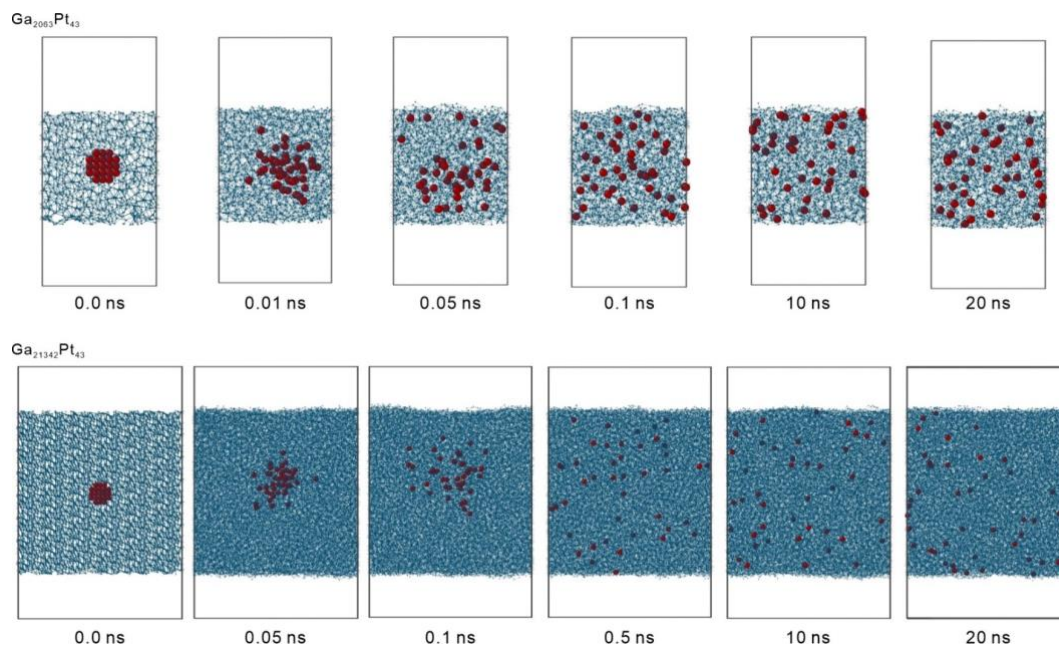


Fig. R14 | The 20 ns NVT molecular dynamics simulations at 923 K on the $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures using the trained Ga–Pt machine-learning potential. Structural snapshots at different time points are shown with the side (xz) view. Dark blue and dark red spheres represent Ga and Pt atoms, respectively, and to better illustrate the single-atom state of Pt atoms, the size of Ga atoms has been relatively reduced.

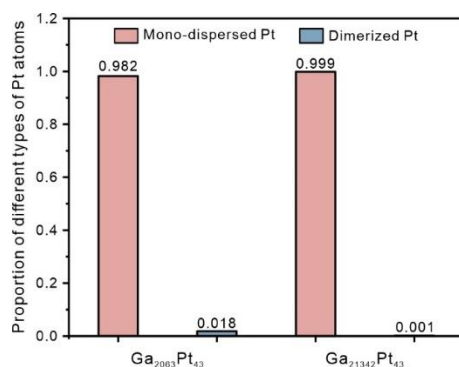
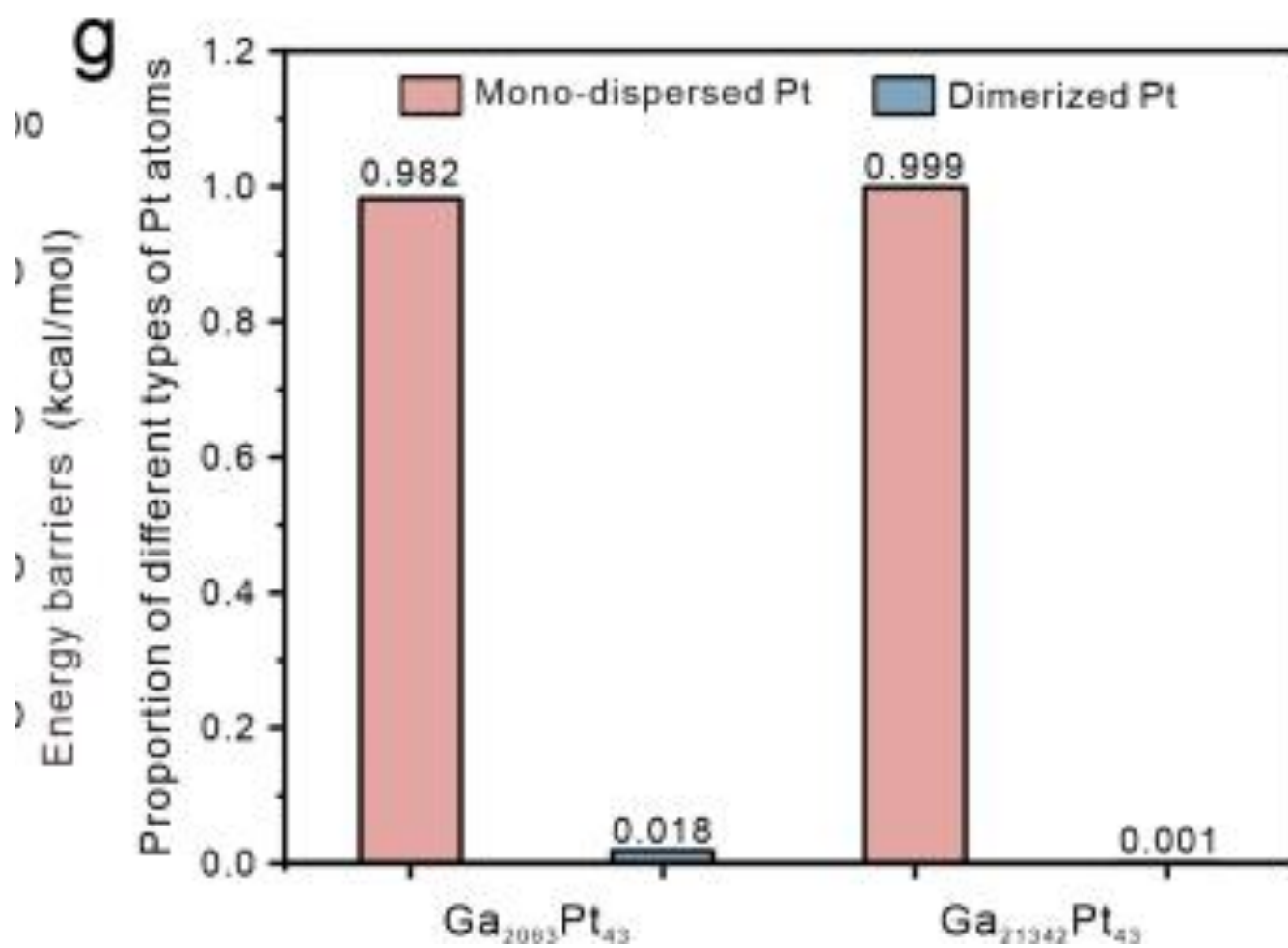


Fig. R15 | The calculated proportions of mono-dispersed and dimerized Pt states in $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{21342}\text{Pt}_{43}$ structures. Mono-dispersed Pt is defined as Pt atoms surrounded exclusively by Ga atoms within a 3.2 Å range, while Dimerized Pt is defined as Pt atom that, within a 3.2 Å range, is surrounded by Ga atoms and exactly one additional Pt atom.



g The calculated proportions of mono-dispersed and dimerized Pt states in $\text{Ga}_{2063}\text{Pt}_{43}$ and $\text{Ga}_{1031}\text{Pt}_{21}$ (Mono-dispersed Pt is defined as Pt atoms surrounded exclusively by Ga atoms within a 3.2 Å range; Dimerized Pt is defined as Pt atom that, within a 3.2 Å range, is surrounded by Ga atoms and an additional Pt atom).

Ga₂₁₃₄₂Pt₄₃ structures (Mono-dispersed Pt is defined as Pt atoms surrounded exclusively by Ga atoms within a 3.2 Å range, while Dimerized Pt is defined as Pt atom that, within a 3.2 Å range, is surrounded by Ga atoms and exactly one additional Pt atom).